Atharva Atul Rege

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Education

National Institute of Technology Karnataka, Surathkal — Mangaluru, India
*Bachelor of Technology in **Computer Science and Engineering** – **8.89 CGPA*** — *Aug 2023 – Present*

Private International English School — Abu Dhabi, UAE
*CBSE Class 12 – **97.6/100*** — *Aug 2022 – Mar 2023*

Experience

I3D Lab, Indian Institute of Science (IISc) — May 2025 – July 2025
Research Intern — *Bengaluru, India*

- Developed a two-stage diffusion-based image generative pipeline that integrates material attribute editing and semantic object placement for image augmentation, leading to a research paper currently under review.
- Implemented a **Material Property Editing** module that uses interpolation in the CLIP embeddings space to alter foreground object properties such as color, transparency, metallic nature, albedo, glow, and surface texture, to generate diverse realistic visual appearances of the object.
- Trained a **Background Context-driven Object Placement** framework that infers the most semantically appropriate 3D position, orientation, and scale for an object to be placed within a given background scene, followed by seamless inpainting of the region to place the object.

Projects

The Brush of Spells : Text-Guided Image Inpainting | *Python, PyTorch* — Nov 2024 – Apr 2025

- Implemented *MMFL: Multimodal Fusion Learning for Text-Guided Image Inpainting* using Generative Adversarial Networks (GANs), leveraging a multimodal architecture to inpaint images based on textual prompts.
- Enhanced the base implementation by adding a CLIP-based loss and Wasserstein GAN (WGAN)-based loss to improve semantic alignment.
- Trained the model on the CUB-200-2011 image-text dataset for better inpainting accuracy for images of birds.

Vision Kinect | *Python, OpenCV, TensorFlow, Keras* — Feb 2024 – May 2024

- Implemented a hand gesture recognition application using YOLO object detection algorithm and mapped each hand gesture to a specific in-game function or control in an interactive Tetris game.
- Enabled fully touchless gameplay using gesture-based controls, demonstrating applications in interactive gaming and Human-Computer Interaction (HCI).

FinBot-AI | *Python, LangChain, Streamlit* — Sep 2024 – Nov 2024

- Developed a Generative AI-powered chatbot for finance-related queries, leveraging OpenAI's GPT-4, GPT-4-Turbo, and GPT-4o APIs to deliver accurate, context-aware responses.
- Utilized the LangChain framework for efficient text generation and processing by developing a Retrieval-Augmented Generation (RAG)-based application with an interactive web interface.

Technical Skills

Languages: Python, C, C++, HTML, CSS
Libraries: PyTorch, TensorFlow, OpenCV, Numpy, Pandas
Frameworks: FastAPI, LangChain, Streamlit, scikit-learn
Technologies: MySQL, Docker, Git, Linux

Extracurricular Activities

IEEE - NITK Student Chapter — Jan 2024 – Present
Executive Member — *Mangaluru, India*

- Contributed to the successful organization of workshops, interactive sessions, and competitions, attracting nationwide participation.
- Mentored a student-led project titled **QuantumQuill: A Transformer-based Sci-Fi Storyteller** focusing on fine-tuning open-source Large Language Models (**LLMs**) for the generation of coherent and creative science fiction stories.